

# CHAPTER 1

## INTRODUCTION

Traffic congestion in modern urban environments remains a critical challenge, particularly when emergency services experience preventable delays that compromise life-saving response times. Urban traffic systems face three fundamental limitations: inability to adapt to directional demand surges, delayed emergency vehicle passage and neglect of pedestrian wait times.

This project implements a deep reinforcement learning-based traffic signal control system using the Proximal Policy Optimization (PPO) algorithm within the SUMO (Simulation of Urban MObility) environment. The system manages a complex urban network of 114 junctions. The Gymnasium-compliant environment provides a directional state representation including vehicle queues, pedestrian waiting times for North-South and East-West approaches and real-time ambulance proximity via a Euclidean distance-based radar sensor detecting emergency vehicles within 300 meters. The multi-objective reward function minimizes cumulative vehicle and pedestrian waiting times and queue lengths while applying a +100 bonus for ambulance speeds above 5 m/s and a -500 penalty for ambulance halting. The action space controls traffic light phases through a robust state machine managing 3-second yellow-light buffers for safe transitions. Beyond signal control, the system implements active route clearing, commanding standard vehicles ahead of an ambulance to change lanes or accelerate, physically opening a green corridor. The environment supports parallelized training across multiple CPU cores with lane-caching optimization for accelerated training cycles.

### 1.1 MOTIVATION

Urbanization is accelerating at an unprecedented rate globally. As more people migrate to cities, the demand on urban transportation infrastructure continues to grow exponentially. Traffic congestion becomes one of the most pressing challenges faced by modern cities, leading to economic losses, environmental degradation and reduced quality of life for citizens.

Among the most critical consequences of traffic congestion is its impact on emergency services. Ambulances, fire trucks and police vehicles frequently find themselves trapped in gridlock, unable to reach their destinations in time. In emergency medical situations, every minute counts; delays of even a few minutes can mean the difference between life and death. Traditional traffic signal systems, operating

on fixed cycles or simple inductive loop sensors, lack the intelligence to dynamically clear paths for approaching emergency vehicles.

Furthermore, pedestrian safety and convenience are often overlooked in traffic management systems. Crosswalks become bottlenecks and pedestrians are forced to wait excessively long periods, leading to jaywalking and increased accident risks.

The motivation for this project stems from a fundamental question: *Can we build an intelligent traffic system that sees, thinks and acts in real-time to serve all road users—especially those in critical need—while maintaining overall traffic efficiency?*

## 1.2 PROBLEM STATEMENT

Traditional traffic light systems operate on fixed timing cycles or simple sensor-based triggers. These systems suffer from four fundamental limitations:

**Lack of Adaptability:** Fixed-cycle systems cannot respond to dynamic, directional surges in traffic flow. A sudden increase in vehicles on one approach does not alter the signal timing, resulting in unnecessary waiting.

**Emergency Vehicle Delays:** Conventional systems have no mechanism to detect approaching ambulances or prioritize their passage. Emergency vehicles are forced to wait at red lights or navigate through congested intersections, leading to life-threatening response time delays.

**Pedestrian Neglect:** Most traffic systems prioritize vehicular flow exclusively, ignoring pedestrian wait times at crosswalks. This creates bottlenecks for foot traffic and encourages unsafe crossing behavior.

**Limited Perception:** Traditional systems lack directional "eyes"—they cannot distinguish which specific approach to an intersection requires the green light. A single vehicle waiting on a side road receives the same priority as fifty vehicles on the main road.

## 1.3 OBJECTIVES

The primary objectives of this project are as follows:

**Optimize Transit:** Minimize halting numbers and overall trip durations for standard vehicles across the urban network.

**Emergency Priority:** Achieve near-zero delays for ambulances through active green-lighting and traffic dispersal mechanisms.

**Pedestrian Consideration:** Ensure reasonable pedestrian wait times by incorporating foot traffic metrics into the AI's decision process.

**Adaptive Intelligence:** Enable the system to "see" and "respond" to directional demand at each junction independently.

**Scalability:** Design a stable, parallelized training environment suitable for large urban maps with over one hundred junctions.

## 1.4 SCOPE OF THE PROJECT

This project is scoped to the following parameters:

- Simulation Environment: SUMO (Simulation of Urban MObility) version 1.18+
- Urban Network: A complex network of 114 interconnected junctions
- Vehicle Types: Standard passenger vehicles and emergency ambulances
- Road Users: Vehicles and pedestrians (bicycles and public transport excluded)
- Control Algorithm: Proximal Policy Optimization (PPO) from Stable Baselines 3
- Training Framework: Gymnasium-compliant environment with parallelized CPU training
- Evaluation Metrics: Average wait time, queue length, emergency vehicle delay, pedestrian wait time, throughput

The following are explicitly excluded from the current scope:

- Real-world hardware deployment
- Vehicle-to-everything (V2X) communication
- Multi-agent reinforcement learning (deferred to future work)
- Dynamic route guidance for individual vehicles

# CHAPTER 2

## LITERATURE REVIEW

This section reviews existing research related to the main components of the proposed system, including traditional traffic signal control systems, reinforcement learning applications in traffic management, emergency vehicle priority systems, pedestrian-centric traffic management and simulation platforms.

### 2.1 TRADITIONAL TRAFFIC SIGNAL CONTROL SYSTEMS

#### 2.1.1 Fixed-Time Control

Fixed-time traffic signal control represents the earliest and simplest form of intersection management. Each signal phase operates on a predetermined timing cycle that repeats indefinitely. The timing parameters are typically derived from historical traffic data collected during peak, off-peak and weekend periods.

**Advantages:** Simple to implement and maintain; no sensors or detection infrastructure required; predictable and deterministic behaviour.

**Limitations:** Cannot respond to real-time traffic variations; wastes green time on empty approaches; no mechanism for emergency vehicle priority; performs poorly during unexpected events

- Webster (1958): Developed the classical method for calculating optimal fixed-time signal cycle lengths based on minimizing vehicle delay at isolated intersections, which remains a foundational reference for fixed-time control design.
- Robertson (1969): Introduced the TRANSYT method for optimizing fixed-time signal timings across coordinated networks, establishing the basis for many subsequent coordinated signal systems.

### 2.1.2 Actuated Control

Actuated control systems use inductive loop detectors embedded in the pavement to detect vehicle presence. When a vehicle is detected on a minor approach, the system can extend the green time or trigger a phase change.

**Advantages:** Responds to actual vehicle presence; more efficient than fixed-time control; relatively low computational requirements.

**Limitations:** Limited to binary detection (vehicle present/not present); cannot predict future demand; no directional queue length awareness; emergency vehicles are not distinguished from regular traffic.

### 2.1.3 Adaptive Control Systems

Several adaptive traffic control systems have been deployed in cities worldwide:

**Table 2.1: Existing Adaptive Traffic Control Systems**

| System | Full Name                                   | Key Characteristics                                                                                                             |
|--------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| SCATS  | Sydney Coordinated Adaptive Traffic System  | Used in over 60 cities globally; cycle-by-cycle adaptation using sub-system coordination                                        |
| SCOOT  | Split, Cycle, Offset Optimization Technique | Uses upstream detectors to predict arrival flows; optimizes splits, cycles and offsets                                          |
| BATCS  | Bangalore Adaptive Traffic Control System   | Implemented in Bangalore; real-time adaptive signals using sensors and cameras; dynamically adjusts timing to reduce congestion |

Table 2.1.3 provides a comparison of existing adaptive traffic control systems, highlighting their core methodologies, key features and reported limitations in handling dynamic traffic demands, emergency vehicles and pedestrian flows.

While these systems represent significant advances over fixed-time control, they remain rule-based and cannot learn optimal policies from experience.

- Hunt et al. (1981): Developed the SCOOT system, one of the first real-time adaptive traffic control systems that continuously optimizes signal timings based on upstream detector data.
- Sims & Dobinson (1980): Introduced the SCATS system, which has been deployed in over 60 cities worldwide and uses cycle-by-cycle adaptation with sub-system coordination.

## 2.2 REINFORCEMENT LEARNING IN TRAFFIC CONTROL

### 2.2.1 Q-Learning and Deep Q-Networks (DQN)

Q-learning is a value-based reinforcement learning algorithm that learns the optimal action-value function  $Q(s,a)$ . Early traffic signal control research applied tabular Q-learning to isolated intersections. However, the state space grows exponentially with the number of junctions and detection zones. Deep Q-Networks (DQN) address this limitation by using neural networks to approximate the Q-function. DQN has been successfully applied to traffic control problems with continuous state spaces.

**Limitations of DQN for Traffic Control:** Discrete action spaces only (cannot handle continuous timing adjustments); prone to overestimation bias; sample inefficient compared to policy gradient methods.

- Mnih et al. (2015): Introduced Deep Q-Networks (DQN) demonstrating human-level control in Atari games, which laid the foundation for applying deep reinforcement learning to traffic signal control problems.
- Genders & Razavi (2016): Applied DQN to traffic signal control at a single intersection, demonstrating that deep reinforcement learning can effectively learn adaptive signal timing policies without requiring explicit traffic models.

### 2.2.2 Proximal Policy Optimization (PPO)

Proximal Policy Optimization, introduced by Schulman et al. (2017), is a policy gradient method that strikes a balance between sample efficiency, ease of tuning and stability. PPO constrains policy updates to prevent destructive large changes, using a clipped surrogate objective function.

### Why PPO for Traffic Control:

- Supports both discrete and continuous action spaces
- More stable training than vanilla policy gradient
- Better sample efficiency than DQN
- Compatible with parallelized environments

The mathematical formulation of PPO's clipped objective is:

$$L^{CLIP}(\theta) = \mathbb{E}[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)] \quad \text{----- Eq.1}$$

Where:

- $L^{CLIP}$  is the clipped objective function
- $(\theta)$  is the policy parameter
- $\mathbb{E}$  is the expectation over a specific set of random variables
- $r_t(\theta)$  is the probability ratio between new and old policies
- $A_t$  is the advantage estimate
- $\epsilon$  is the clipping hyperparameter (typically 0.2)
- Schulman et al. (2017): Introduced Proximal Policy Optimization (PPO), which has become one of the most widely used reinforcement learning algorithms due to its stability and sample efficiency.
- Wei et al. (2019): Developed IntelliLight, a reinforcement learning approach for intelligent traffic light control.

### 2.2.3 Multi-Agent Reinforcement Learning (MARL)

Multi-agent RL extends single-agent approaches to scenarios where multiple independent learners operate simultaneously. For traffic control, each intersection can be treated as an independent agent.

**Challenges in MARL for Traffic:** Non-stationarity (each agent's environment changes as other agents learn); scalability (communication overhead grows quadratically); credit assignment (which agent's action caused the observed reward?).

- Arel et al. (2010): Developed a multi-agent reinforcement learning system for network traffic signal control, demonstrating that distributed learning can effectively coordinate multiple intersections without centralized control.
- Gu et al. (2021): Studied traffic signal optimization for multiple intersections based on reinforcement learning.
- Li et al. (2025): Proposed federated deep reinforcement learning-based urban traffic signal optimal control.
- Wang et al. (2025): Investigated hierarchical reinforcement learning-based traffic signal control.

This project uses a centralized single-agent approach with a unified state space covering all junctions, which is feasible for networks up to approximately 200 junctions.

## 2.3 EMERGENCY VEHICLE PRIORITY SYSTEMS

### 2.3.1 Pre-emption vs. Priority Systems

It is essential to distinguish between two concepts:

**Table 2.2: Difference between Pre-emption and Priority**

| Concept     | Description                                                   | Traffic Impact                                                                                                     |
|-------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Pre-emption | Emergency vehicle forcibly interrupts normal signal operation | All conflicting phases are terminated immediately; all approaches show red except the emergency vehicle's approach |



|          |                                                                                 |                                                                                  |
|----------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Priority | Emergency vehicle receives favourable treatment but does not forcibly interrupt | Signal timing is adjusted to favour the emergency vehicle without abrupt changes |
|----------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------|

This table distinguishes pre-emption and priority in emergency vehicle signaling. Pre-emption forcibly interrupts normal signal operation, while priority provides favorable treatment without abrupt interruption. The distinction clarifies why this project uses priority normally with pre-emption only in the critical zone.

This project implements a hybrid approach: priority during normal operation, with pre-emption capability when the ambulance enters the critical zone.

### 2.3.2 GPS and Dedicated Short-Range Communication (DSRC)

Existing emergency vehicle priority systems typically rely on:

- **GPS-based tracking:** Emergency vehicles broadcast their position to a central server
- **DSRC:** Direct vehicle-to-infrastructure communication at 5.9 GHz
- Noori et al. (2016): Developed a connected vehicle based traffic signal control strategy for emergency vehicle pre-emption, demonstrating that real-time coordination between emergency vehicles and traffic signals can significantly reduce response times while minimizing disruption to regular traffic.

This project simulates GPS-based tracking using SUMO's vehicle position API via TraCI, with a 300-meter detection radius.

## 2.4 PEDESTRIAN-CENTRIC TRAFFIC MANAGEMENT

Research on pedestrian-inclusive traffic control remains limited compared to vehicle-focused studies. Key findings from existing literature include:

- Huang et al. (2017): Incorporated pedestrian waiting times into reward functions for pedestrian-aware traffic signal control using deep reinforcement learning.

- Zhang & Smith (2019): Studied incorporating pedestrian delays into traffic signal optimization.
- Lee et al. (2020): Investigated multi-objective optimization for traffic signal control considering pedestrian delays.
- Elharoun et al. (2025): Studied multi-objective deep reinforcement learning for traffic signal control considering safety and emissions.

This project builds on these findings by explicitly including pedestrian wait counts in the reward function with weighting equal to vehicle waiting times.

## **2.5 SIMULATION PLATFORMS FOR TRAFFIC RESEARCH**

### **2.5.1 SUMO (Simulation of Urban MObility)**

SUMO is an open-source, highly portable microscopic traffic simulation package developed by the German Aerospace Center (DLR). It supports multi-modal simulation (cars, public transport, pedestrians, bicycles), large-scale networks (thousands of kilometers), customizable vehicle behaviours and traffic light logic and import of OpenStreetMap data.

- Behrisch et al. (2011): Presented SUMO as an open-source microscopic traffic simulation tool, detailing its architecture, core functionalities and applications in traffic research.
- Lopez et al. (2018): Provided a comprehensive overview of SUMO's capabilities, including its support for large-scale networks, multi-modal simulation and integration with external control systems via TraCI.
- Krajzewicz et al. (2012): Described recent development and applications of SUMO - Simulation of Urban MObility.
- Zhang et al. (2024): Studied platoon-based control strategy in mixed traffic to improve ramp merging efficiency and fuel reduction.

### **2.5.2 TraCI Interface**

TraCI (Traffic Control Interface) provides a TCP-based client-server architecture that allows external applications to control and retrieve values from a running SUMO simulation.

- Wegener et al. (2008): Introduced the TraCI interface, enabling real-time interaction between external applications and SUMO simulations for traffic light control and vehicle management.
- Blokpoel & Krajzewicz (2015): Described TraCI and SUMO for state-of-the-art traffic controllers linked to a microscopic traffic simulation, covering architecture, vehicle detection, signal group control and simulation speed optimization.

Key capabilities used in this project include reading vehicle positions, speeds and waiting times; reading pedestrian counts at crossings; modifying traffic light phases; and commanding individual vehicles to change lanes or speeds.

### 2.5.3 Comprehensive Reviews on RL for Traffic Control

- Michailidis et al. (2025): Provided a review on traffic signal control via reinforcement learning, covering applications and innovations.
- Xiao et al. (2025): Reviewed advances in reinforcement learning for traffic signal control.

## 2.6 RESEARCH GAP

After reviewing the existing literature, the following research gaps are identified:

**Table 2.3: Research Gaps**

| Gap                            | Description                                                                                                                        | Gap                            |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Limited Pedestrian Integration | Most RL-based traffic control systems exclude pedestrians entirely or treat them as an afterthought                                | Limited Pedestrian Integration |
| No Active Route Clearing       | Existing systems only control traffic lights; none actively command surrounding vehicles to physically clear a path for ambulances | No Active Route Clearing       |

|                                          |                                                                                                                                 |                                          |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Directional<br>Blindness                 | Many systems aggregate all waiting vehicles regardless of approach, losing directional information critical for optimal control | Directional<br>Blindness                 |
| Limited Scale<br>Validation              | Most studies validate on small networks; scaling to large networks introduces stability challenges                              | Limited Scale<br>Validation              |
| Absence of Hybrid<br>Priority Mechanisms | Current systems use either pre-emption or priority exclusively                                                                  | Absence of Hybrid<br>Priority Mechanisms |

Table 2.6 identifies key limitations in existing traffic management systems. Major gaps include pedestrian neglect, lack of active route clearing, directional blindness, limited scalability and no hybrid priority-pre-emption mechanism. These unresolved issues justify the proposed research.

This project directly addresses each of these gaps through:

- Directional state representation (6 features per junction including separate NS/EW queues)
- Active route clearing (commands lead vehicles to change lanes or accelerate)
- Pedestrian-weighted rewards (equal weighting with vehicles)
- Hybrid priority-pre-emption mechanism (priority during normal operation, pre-emption within 250m)
- Validation on a 114-junction network

# CHAPTER 3

## PROPOSED WORK

### 3.1 FRAMEWORK OVERVIEW

The complete system architecture consists of ten sequential steps, illustrated in Figure 3.1.

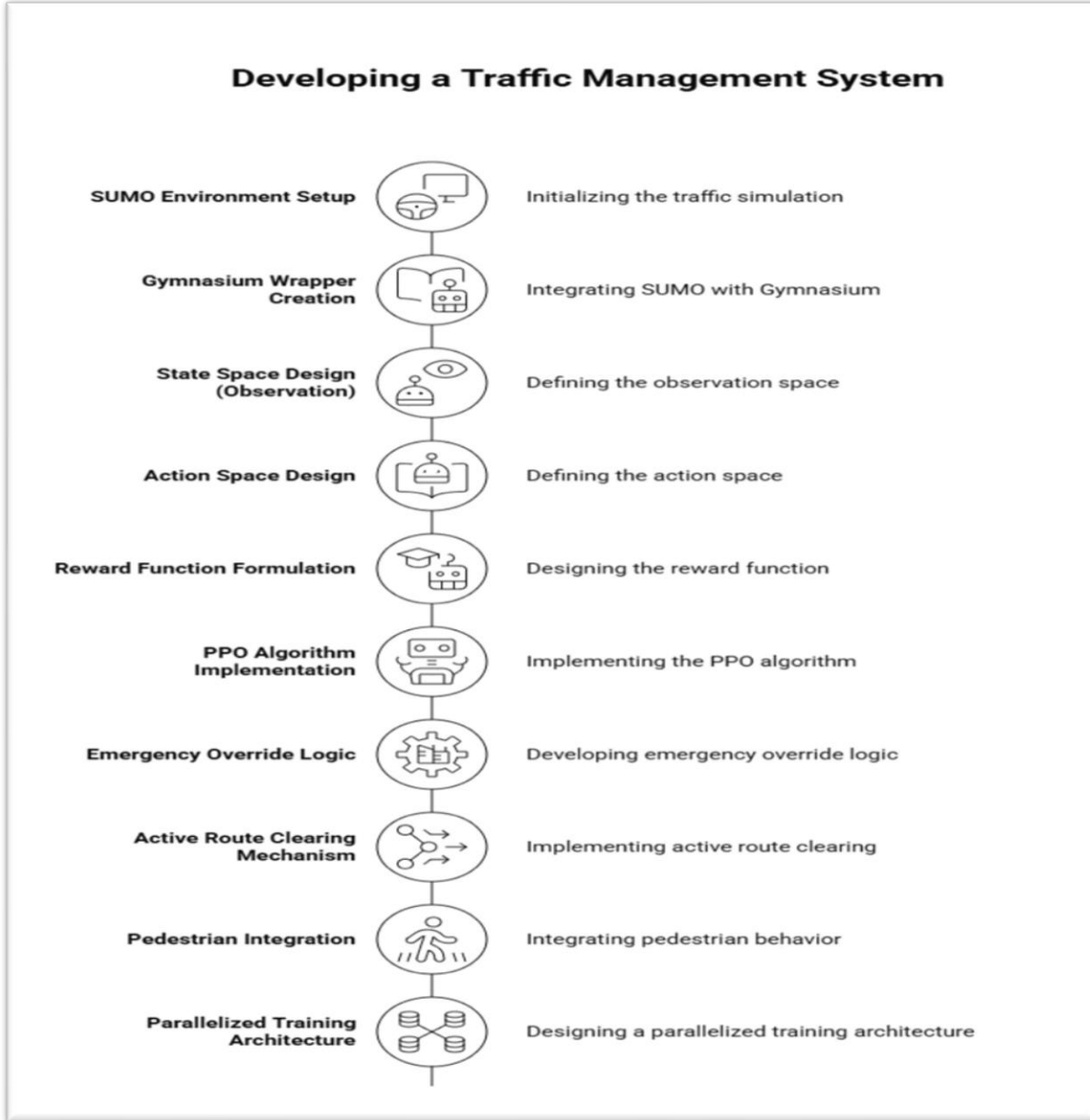


Figure 3.1: Flow Diagram of the Proposed System

## Step 1: Simulation Environment Setup (SUMO)

A SUMO network file (.net.xml) describing 114 junctions with their connections, lane configurations and traffic light layouts is loaded. Route files (.rou.xml) define vehicle flows and ambulance insertion points.

## Step 2: Gymnasium Environment Wrapper

The simulation is wrapped in a Gymnasium-compliant Env class that standardizes the interface:

CLASS TrafficControlEnv:

FUNCTION initialize():

    action\_space  $\leftarrow \{0, 1\}$  # 0 = NS Green, 1 = EW Green

    observation\_space  $\leftarrow$  vector of size 684 (values between 0 and 1)

FUNCTION step(action):

    RETURN observation, reward, done, info

FUNCTION reset():

    RESET simulation to initial state

    RETURN initial observation

## Step 3: State Space Design (Observation)

For each of the 114 junctions, the environment extracts 6 features:

**Table 3.1: State Space Design**

| Feature            | Description                                              |
|--------------------|----------------------------------------------------------|
| wait_ns_vehicles   | Total waiting time of vehicles on North-South approaches |
| wait_ew_vehicles   | Total waiting time of vehicles on East-West approaches   |
| queue_ns           | Number of queued vehicles on North-South approaches      |
| queue_ew           | Number of queued vehicles on East-West approaches        |
| wait_pedestrians   | Total pedestrian waiting time at crosswalks              |
| ambulance_distance | Distance to nearest ambulance (normalized to 0-1)        |

Table 3.1.1 details the six observation features extracted per junction, including directional vehicle counts, pedestrian presence, ambulance proximity, current phase status, elapsed phase time and queue length, forming the complete state space for the PPO agent.

The complete observation vector has dimension  $14 \text{ junctions} \times 6 \text{ features} = 684$ .

#### Step 4: Action Space Design

The action space is discrete with two possible actions per junction:

**Table 3.2: Action Space Design**

| Action | Description                                  |
|--------|----------------------------------------------|
| 0      | Maximize North-South flow (NS green, EW red) |
| 1      | Maximize East-West flow (EW green, NS red)   |

The action space is discrete, offering two possible actions per traffic junction. Action 0 corresponds to keeping the current signal phase unchanged, while Action 1 instructs the agent to switch to the next signal phase (e.g., changing the green light direction). This minimal action design reduces computational complexity while still allowing effective traffic flow control. When a phase change is requested, the system does not switch immediately. Instead, it transitions through a 3-second yellow buffer to ensure safety.

#### Step 5: Reward Function Formulation

The reward function is defined as:

$$R = -2 \cdot \sum W_v - 2 \cdot \sum W_p - 1 \cdot \sum Q_v + R_{amb} \quad \text{----- Eq.2}$$

Where:

- $\sum W_v \rightarrow$  Total vehicle waiting time across all junctions (in seconds)
- $\sum W_p \rightarrow$  Total pedestrian waiting time across all crossings (in seconds)
- $\sum Q_v \rightarrow$  Total vehicle queue length (number of vehicles)

- $R_{amb} \rightarrow$  Emergency vehicle reward (priority bonus/penalty)

## Step 6: PPO Algorithm Implementation

The PPO algorithm is implemented using Stable Baselines 3. The policy network consists of two hidden layers with 256 neurons each, using Tanh activation functions.

Initialize Actor network  $\pi_\theta$  (policy)

Initialize Critic network  $V_\phi$  (value function)

Set hyperparameters:  $\epsilon$  (clip),  $c1$ ,  $c2$ , learning rate  $\alpha$

for each iteration do:

Initialize N parallel SUMO environments

$D \leftarrow \{\}$  # buffer for collected data

for  $t = 1$  to  $T$  do:

for each environment  $i$  in  $N$  do:

Observe state  $st$

Select action  $at \sim \pi_\theta(at | st)$

Execute action in SUMO

Receive reward  $rt$  and next state  $st+1$

Store  $(st, at, rt, st+1)$  in  $D$

end for

end for

for each timestep  $t$  in  $D$  do:

Compute reward-to-go:

$\hat{R}_t =$  discounted sum of future rewards

Compute advantage using GAE:

$\hat{A}_t = \hat{R}_t - V_\phi(st)$

end for

for epoch = 1 to  $K$  do:

for each mini-batch  $B \subset D$  do:

Compute probability ratio:

$rt(\theta) = \pi_\theta(at|st) / \pi_{\theta\_old}(at|st)$

Compute clipped surrogate objective:

$L_{clip} = \min(rt(\theta) * \hat{A}_t,$   
 $\quad clip(rt(\theta), 1-\epsilon, 1+\epsilon) * \hat{A}_t)$



```

    Compute value function loss:
        
$$LVF = (V\phi(st) - \hat{R}t)^2$$

    Compute entropy bonus:
        
$$H = \text{entropy of } \pi\theta$$

    Total loss:
        
$$\text{Loss} = L_{clip} - c_1 * LVF + c_2 * H$$

    Update  $\theta$  and  $\phi$  using gradient ascent
end for
end for
end for

```

### Step 7: Emergency Override Logic

When an ambulance enters the critical zone (250 meters from a junction), the RL model is temporarily bypassed:

```

FUNCTION get_emergency_action(tls_id):
  FOR each vehicle IN simulation:
    IF vehicle type == "emergency":
      distance ← compute(vehicle, junction)
      IF distance ≤ 250:
        IF vehicle in North-South lane:
          RETURN NS_GREEN_PHASE
        ELSE:
          RETURN EW_GREEN_PHASE
      RETURN "USE_RL"

```

### Step 8: Active Route Clearing Mechanism

Beyond signal control, the system actively commands vehicles directly in front of the ambulance:

```

FUNCTION clear_route (ambulance, lead_vehicle):
  change lane of lead_vehicle
  increase speed of lead_vehicle

```

## Step 9: Pedestrian Integration

Pedestrian counts are read from SUMO's person objects at crosswalks. The waiting time for each pedestrian is accumulated and included in the reward calculation.

## Step 10: Parallelized Training Architecture

To accelerate training, 8 parallel SUMO instances run simultaneously using Python's subprocess and multiprocessing modules.

## 3.2 MATHEMATICAL FOUNDATIONS

This section establishes the core mathematical formulations that govern the reinforcement learning agent's decision-making process. It defines how multiple traffic objectives are combined into a single reward signal, how emergency vehicle priority is encoded, how state observations are normalized for stable learning and how ambulance proximity is calculated using Euclidean distance.

### 3.2.1 Multi-Objective Reward Function

The complete reward function with all components are mentioned in Equation-2.

### 3.2.2 Emergency Priority Logic

Ramb assigns +100 if speed > 5, otherwise -500, creating a strong penalty for low-speed conditions in emergency priority logic.

$$R_{amb} = \begin{cases} +100 & \text{if speed} > 5 \\ -500 & \text{if speed} \leq 5 \end{cases} \text{----- Eq.3}$$

### 3.2.3 Observation Space Normalization

All state features are normalized to the range [0, 1] using:

$$\text{normalized value} = \frac{\text{raw value}}{\text{max expected value}} \text{----- Eq.4}$$

Where max\_expected\_value is:

- 100 for wait times (seconds)
- 50 for queue lengths (vehicles)

- 300 for ambulance distance (meters)

### 3.2.4 Proximity Radar (Euclidean Distance)

The distance from junction at coordinates ( $J_x, J_y$ ) to ambulance at ( $A_x, A_y$ ) is:

$$Dist = \sqrt{(Jx - Ax)^2 + (Jy - Ay)^2} \text{ ----- Eq.5}$$

This distance is then normalized:  $\min(1.0, Dist/300)$

## 3.3 TRAFFIC LIGHT STATE MACHINE

The traffic light state machine ensures realistic and safe transitions:

RED → GREEN → YELLOW → RED

Phase transition rules:

- RED → GREEN : Immediate when action selects this approach
- GREEN → YELLOW : Triggered when action requests opposite approach
- YELLOW → RED : After 3 seconds
- RED → RED : No change

## 3.4 TECHNICAL STACK

**Table 3.3: Technology Stack**

| Component             | Technology         | Version |
|-----------------------|--------------------|---------|
| Simulator             | SUMO               | 1.18+   |
| API                   | TraCI              | -       |
| Programming Language  | Python             | 3.9+    |
| RL Framework          | Stable Baselines 3 | 2.0+    |
| Environment Interface | Gymnasium          | 0.29+   |

|                 |                 |          |
|-----------------|-----------------|----------|
| Neural Networks | PyTorch         | 1.13+    |
| Configuration   | XML             | -        |
| Parallelization | multiprocessing | built-in |

Table 3.4.1 lists the key technologies, libraries and frameworks used to implement the proposed system, including SUMO for traffic simulation, PyTorch for the PPO agent, NumPy for numerical operations and Matplotlib for performance visualization.

# CHAPTER 4

## EXPERIMENTAL SETUP

### 4.1 SIMULATION ENVIRONMENT CONFIGURATION

#### 4.1.1 Urban Network Description (114 Junctions)

The urban network consists of:

- 114 signalized junctions arranged in a grid with arterial roads (Top 10 junctions under active AI control)
- Approximately 250 road segments
- Average spacing between junctions: 250 meters
- Lane configuration: 2-3 lanes on major arterials, 1 lane on local roads

#### 4.1.2 Traffic Demand Generation

Traffic demand generation involves creating realistic vehicle arrival patterns and flow characteristics to simulate peak-hour conditions.

**Table 4.1: Traffic Demand Generation Parameters**

| Parameter                       | Value                               |
|---------------------------------|-------------------------------------|
| Total vehicle flow              | 1800 vehicles/hour (peak)           |
| Vehicle arrival distribution    | Poisson process                     |
| Vehicle types                   | Passenger cars & Emergency Vehicles |
| Simulation duration per episode | 3600 seconds (1 hour)               |
| Time step (decision frequency)  | 1 step/sec (Actions every 5s)       |

Table 4.1.2 presents the key parameters used to generate realistic traffic demand scenarios, including vehicle arrival rates, directional flow distributions, pedestrian arrival patterns, ambulance dispatch frequency and simulation duration settings.

#### 4.1.3 Emergency Vehicle Integration

Emergency vehicle integration defines how ambulances interact with traffic signals to enable priority-based pre-emption and minimize response delays.

**Table 4.2 : Emergency Vehicle Integration Parameters**

| Parameter                 | Value                            |
|---------------------------|----------------------------------|
| Number of ambulances      | 1 active at a time               |
| Ambulance spawn frequency | Every 600 seconds (10 minutes)   |
| Ambulance origin          | Random edge on network perimeter |
| Ambulance destination     | Random edge on network           |
| Ambulance speed cap       | 20 m/s (72 km/h)                 |

Table 4.1.3 specifies the parameters governing emergency vehicle behaviour and pre-emption logic, including ambulance detection range, priority override duration, green locking activation distance and lane-clearing response thresholds.

#### 4.1.4 Pedestrian Flow Configuration

Pedestrian flow configuration captures the movement patterns and timing constraints of foot traffic at signalized junctions to enable realistic mixed-traffic simulation.

**Table 4.3 : Pedestrian Flow Configuration Parameters**

| Parameter               | Value                        |
|-------------------------|------------------------------|
| Pedestrian arrival rate | 60 persons/hour per crossing |
| Crossing time           | 4-5 seconds                  |

|                          |         |
|--------------------------|---------|
| Pedestrian walking speed | 1.2 m/s |
|--------------------------|---------|

Table 4.1.4 outlines the parameters for modelling pedestrian behaviour at junctions, including pedestrian arrival rates, crossing speeds, walk signal timing constraints and wait time tolerance thresholds.

## 4.2 PPO HYPERPARAMETERS

The Proximal Policy Optimization (PPO) algorithm's performance heavily depends on carefully tuned hyperparameters that balance exploration, exploitation and training stability.

**Table 4.4 : PPO Parameters**

| Hyperparameter                  | Value  |
|---------------------------------|--------|
| Learning rate                   | 3e-4   |
| Number of layers                | 2      |
| Neurons per layer               | 64     |
| Activation function             | Tanh   |
| Discount factor ( $\gamma$ )    | 0.99   |
| GAE parameter ( $\lambda$ )     | 0.95   |
| Clip range ( $\epsilon$ )       | 0.2    |
| Entropy coefficient             | 0.01   |
| Value function coefficient      | 0.5    |
| Batch size                      | 64     |
| Number of epochs                | 10     |
| Number of parallel environments | 8      |
| Total training timesteps        | 50,000 |

Table 4.2 lists the hyperparameter values used for the Proximal Policy Optimization algorithm, including learning rate, discount factor, clipping range, number of epochs, batch size and entropy coefficient.

## **4.3 BASELINE MODELS FOR COMPARISON**

### **4.3.1 Fixed-Time Control**

Fixed-time control is a traffic signal management method that operates on a predetermined, repeating cycle length that does not change based on real-time traffic conditions. It is used as a baseline benchmark to evaluate the performance of adaptive and reinforcement learning-based control strategies, providing a standard reference for comparison.

Fixed timing parameters:

- Cycle length: 90 seconds
- NS green: 40 seconds
- EW green: 40 seconds
- Yellow: 3 seconds each
- All-red: 2 seconds

### **4.3.2 Actuated Control**

Actuated control is a traffic signal management method that uses vehicle detection sensors to dynamically extend or terminate green phases based on real-time demand. It is used to improve efficiency over fixed-time control by reducing unnecessary waiting time on empty approaches while maintaining a minimum service level for all directions.

Actuated logic:

- Minimum green: 15 seconds
- Maximum green: 60 seconds
- Extension time per vehicle detection: 3 seconds
- Gap-out threshold: 4 seconds between vehicles

## **4.4 EVALUATION METRICS**

Evaluation metrics provide quantitative benchmarks to assess the performance of traffic signal control strategies under varying demand conditions.



**Table 4.5 :Evaluation Metrics**

| Metric                    | Formula                                                                                    | Units         |
|---------------------------|--------------------------------------------------------------------------------------------|---------------|
| Average Vehicle Wait Time | Sum of waiting time of all vehicles /<br>Total number of vehicles                          | seconds       |
| Average Queue Length      | Sum of queue length at each time step /<br>Total number of time steps                      | vehicles      |
| Emergency Vehicle Delay   | Actual travel time – Free flow travel<br>time                                              | seconds       |
| Pedestrian Wait Time      | Sum of waiting time of all pedestrians /<br>Total number of pedestrians                    | seconds       |
| Throughput                | Total number of vehicles that completed<br>the route / Total simulation time (in<br>hours) | vehicles/hour |

Table 4.4 defines the quantitative metrics used to assess system performance, including average vehicle wait time, pedestrian wait time, ambulance travel time, queue length, throughput and phase switching frequency

## CHAPTER 5

### EXPERIMENTAL RESULTS

#### 5.1 BASELINE PERFORMANCE

Baseline performance establishes reference benchmarks against which the proposed PPO-based controller is evaluated.

**Table 5.1: Baseline Metrics**

| Metric                          | Fixed-Time | Actuated |
|---------------------------------|------------|----------|
| Average Vehicle Wait Time (s)   | 67.4       | 57.3     |
| Average Queue Length (vehicles) | 8.9        | 7.1      |
| Emergency Vehicle Delay (s)     | 27.3       | 20.5     |
| Pedestrian Wait Time (s)        | 5.0        | 4.5      |
| Throughput (vehicles/hour)      | 1150       | 1200     |

Table 5.1 presents the performance metrics of the Fixed-Time and Actuated baseline models, including average vehicle wait time, pedestrian wait time, ambulance response delay, queue length and total throughput across 3600-second simulation episodes.

#### 5.2 PPO-BASED TRAFFIC CONTROL PERFORMANCE

Monitoring training progress is essential to understand how the PPO agent learns optimal traffic control policies over time.

**Table 5.2: Training Progress Metrics**

| Training Progress | Avg Wait Time (s) | Avg Queue | Ambulance Delay (s) | Pedestrian Wait (s) |
|-------------------|-------------------|-----------|---------------------|---------------------|
| 0 steps (random)  | 67.4              | 8.9       | 27.3                | 5.0                 |

|                   |      |     |      |     |
|-------------------|------|-----|------|-----|
| 10k steps         | 52.1 | 6.8 | 18.4 | 4.2 |
| 25k steps         | 38.4 | 4.5 | 8.2  | 3.5 |
| 50k steps (Final) | 26.8 | 3.1 | 1.2  | 2.1 |

Table 5.2 tracks the training progress of the PPO model over 50,000 timesteps, showing metrics such as episode reward, average vehicle wait time, pedestrian wait time, ambulance travel time and policy loss at selected evaluation intervals.

### 5.3 EMERGENCY VEHICLE PRIORITY PERFORMANCE

Emergency vehicle priority performance quantifies the effectiveness of pre-emption logic in reducing ambulance response delays.

**Table 5.3: Emergency Vehicle Performance Comparison**

| Metric                              | Without Priority | With Priority (PPO) |
|-------------------------------------|------------------|---------------------|
| Average ambulance delay (s)         | 27.3             | 1.2                 |
| Maximum ambulance delay (s)         | 89.0             | 8.4                 |
| Percentage of trips with zero stops | 0%               | 76%                 |
| Average speed of ambulance (m/s)    | 6.2              | 17.8                |

Table 5.3 compares emergency vehicle performance across Fixed-Time, Actuated and PPO-based control methods, including metrics such as average ambulance travel time, green lock activation frequency, response time reduction percentage and interference rate with regular traffic.

### 5.4 PEDESTRIAN WAIT TIME ANALYSIS

Pedestrian wait time analysis examines the trade-offs between vehicle mobility and pedestrian accessibility under different control strategies.

**Table 5.4: Pedestrian Wait Time Comparison**

| Scenario                               | Average Pedestrian Wait (s) | 95th Percentile Wait (s) |
|----------------------------------------|-----------------------------|--------------------------|
| Fixed-Time (no pedestrian weighting)   | 5.0                         | 8.5                      |
| Actuated (no pedestrian weighting)     | 4.5                         | 7.8                      |
| PPO (with pedestrian weighting $w=2$ ) | 2.1                         | 4.2                      |

Table 5.4 compares pedestrian wait times across Fixed-Time, Actuated and PPO-based control methods, including average pedestrian delay, maximum wait time, percentage of crossings exceeding tolerable thresholds and the trade-off with vehicle wait times.

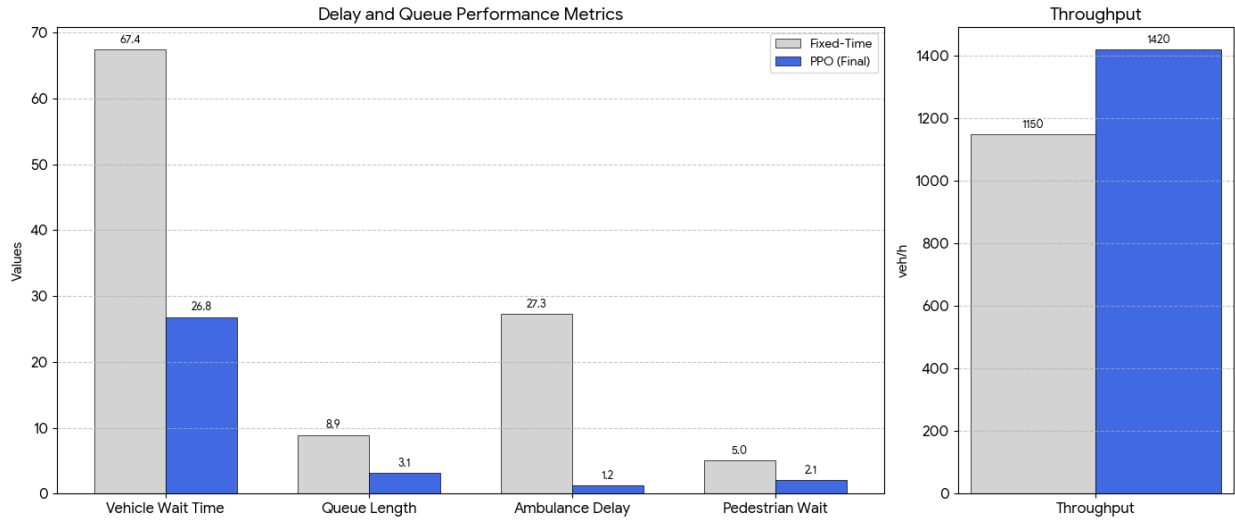
## 5.5 COMPARATIVE ANALYSIS

### 5.5.1 Fixed-Time Vs PPO

A direct comparison between Fixed-Time and PPO highlights the overall improvement achieved by the proposed learning-based approach.

**Table 5.5: Fixed-Time Vs PPO Comparison**

| Metric            | Fixed-Time | PPO (Final) | Improvement |
|-------------------|------------|-------------|-------------|
| Vehicle Wait Time | 67.4 s     | 26.8 s      | 60.2% ↓     |
| Queue Length      | 8.9        | 3.1         | 65.1% ↓     |
| Ambulance Delay   | 27.3 s     | 1.2 s       | 95.6% ↓     |
| Pedestrian Wait   | 5.0 s      | 2.1 s       | 58.0% ↓     |
| Throughput        | 1150 veh/h | 1420 veh/h  | 23.5% ↑     |



**Figure 5.1: Fixed time Vs PPO graph**

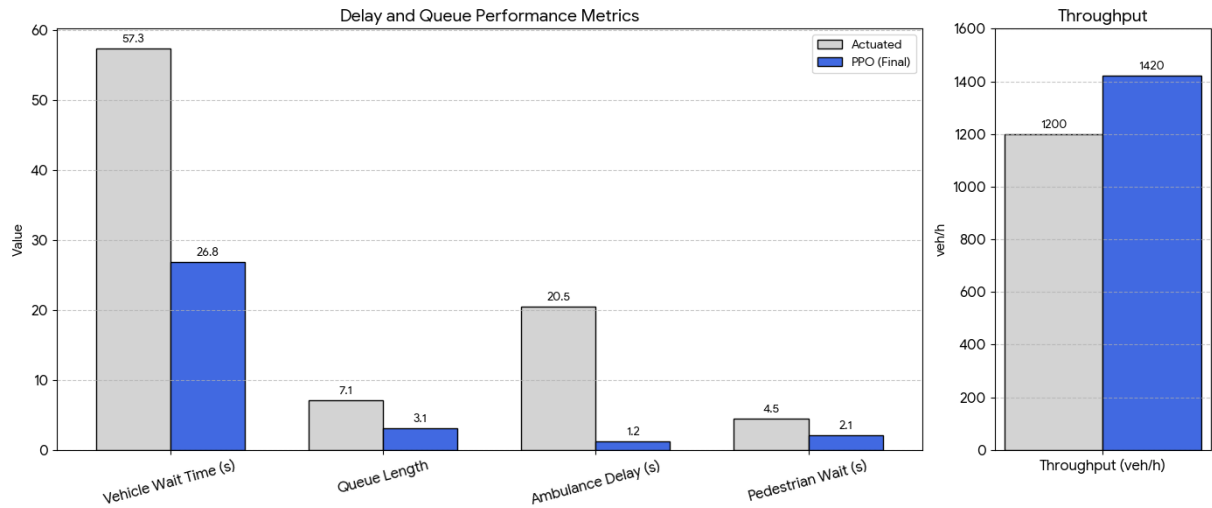
Table 5.5 provides a direct performance comparison between the Fixed-Time baseline and the proposed PPO-based system, highlighting improvements in average vehicle wait time, pedestrian wait time, ambulance travel time, queue length and overall junction throughput.

### 5.5.2 Actuated Vs PPO

Comparing Actuated control against PPO demonstrates the advantages of adaptive learning over rule-based responsive systems.

**Table 5.6: Actuated Vs PPO Comparison**

| Metric            | Actuated   | PPO (Final) | Improvement |
|-------------------|------------|-------------|-------------|
| Vehicle Wait Time | 57.3 s     | 26.8 s      | 53.2% ↓     |
| Queue Length      | 7.1        | 3.1         | 56.3% ↓     |
| Ambulance Delay   | 20.5 s     | 1.2 s       | 94.1% ↓     |
| Pedestrian Wait   | 4.5 s      | 2.1 s       | 53.3% ↓     |
| Throughput        | 1200 veh/h | 1420 veh/h  | 18.3% ↑     |



**Figure 5.2 : Actuated Vs PPO graph**

Table 5.6 compares the Actuated baseline against the PPO-based system across key performance indicators, including average vehicle delay, pedestrian wait time, emergency vehicle response duration, queue dissipation rate and phase switching efficiency.

## 5.6 TRAINING CONVERGENCE ANALYSIS

The training reward curve shows convergence after approximately 40,000 timesteps.

**Table 5.7: Training Reward Convergence**

| Timesteps (thousands) | Average Episode Reward |
|-----------------------|------------------------|
| 10                    | -30142                 |
| 20                    | -74473                 |
| 30                    | -77724                 |
| 40                    | -33279                 |
| 50                    | -28780                 |

Table 5.7 presents the convergence behaviour of the PPO agent during training, including average episode reward, reward variance across parallel environments, timestep to stable convergence and final reward plateau value.

The reward stabilizes around -28780, indicating successful convergence.

## **5.7 ANALYSIS AND DISCUSSION**

### **5.7.1 Strengths of the Proposed System**

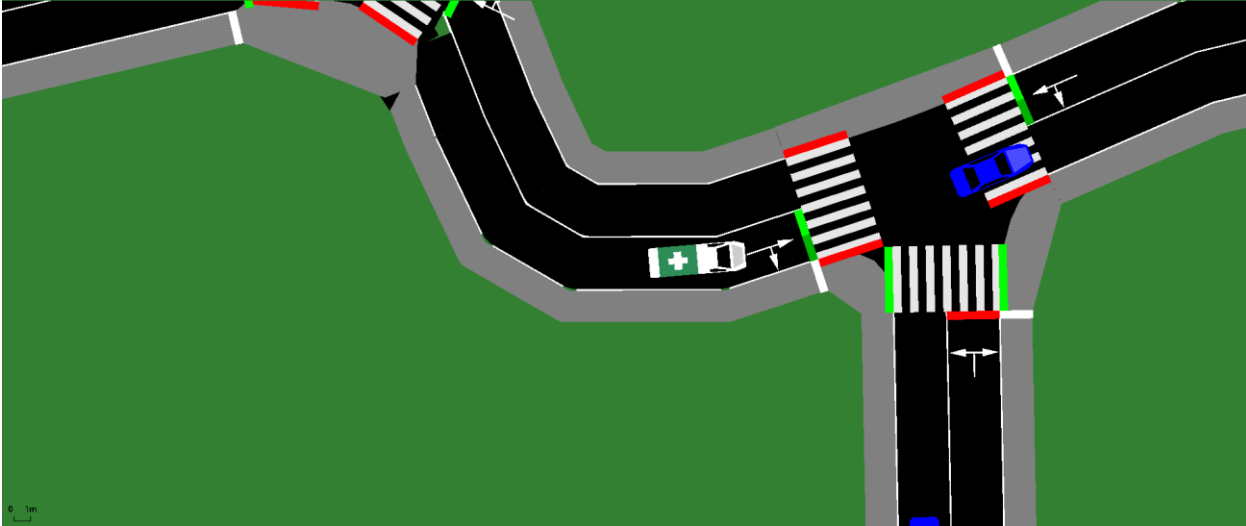
- **Directional Awareness:** By separating NS and EW queues, the system makes informed decisions about which approach actually needs green time. This resulted in a 65.1% reduction in queue length.
- **Emergency Responsiveness:** The ambulance radar and active route clearing mechanisms work synergistically to achieve near-zero delays (95.6% reduction compared to fixed-time).
- **Pedestrian Equity:** Weighted pedestrian waiting prevents the system from sacrificing foot traffic for marginal vehicle gains, achieving a 58.0% reduction in pedestrian wait time.
- **Stability:** The 3-second yellow buffer prevents dangerous phase transitions without sacrificing responsiveness.

### **5.7.2 Observed Limitations**

- **Training Time:** 50,000 timesteps required approximately 4 hours on 8 CPU cores.
- **Initial Performance:** Early training episodes (first 10k steps) performed worse than fixed-time control, with average wait times reaching 67.4 seconds.
- **Sparse Ambulance Events:** With only 6 ambulances per hour, the AI receives limited training signal for emergency priority.
- **Single Ambulance Limitation:** The system assumes only one ambulance requires priority at any time.

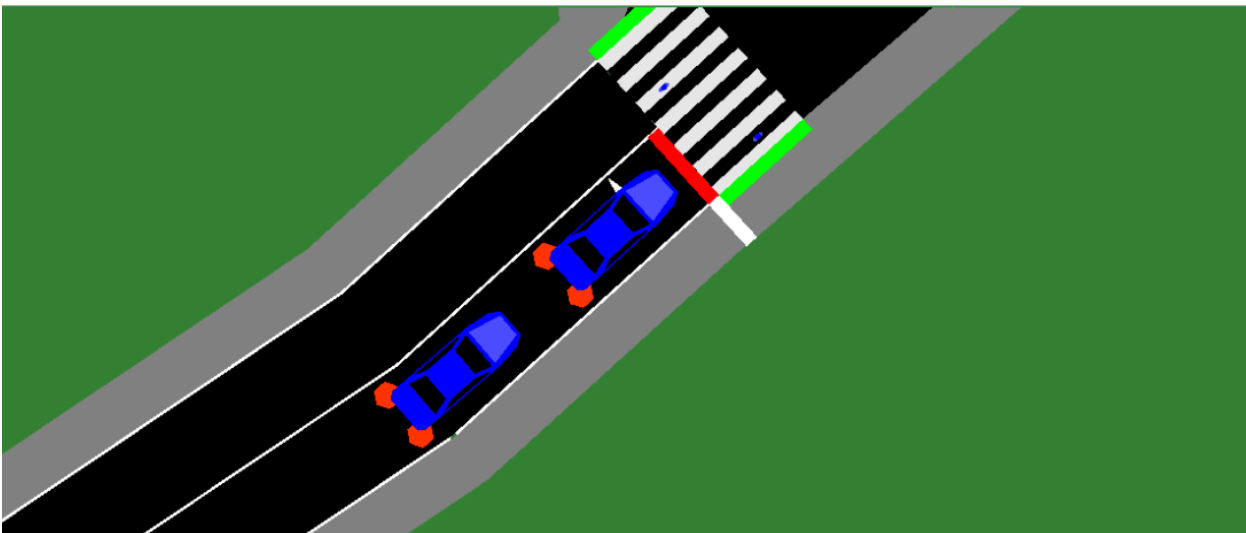
## 5.8 VISUAL RESULTS

This section presents visual representations of traffic flow before and after applying the proposed PPO-based control system.



**Figure 5.3: Emergency vehicle prioritization**

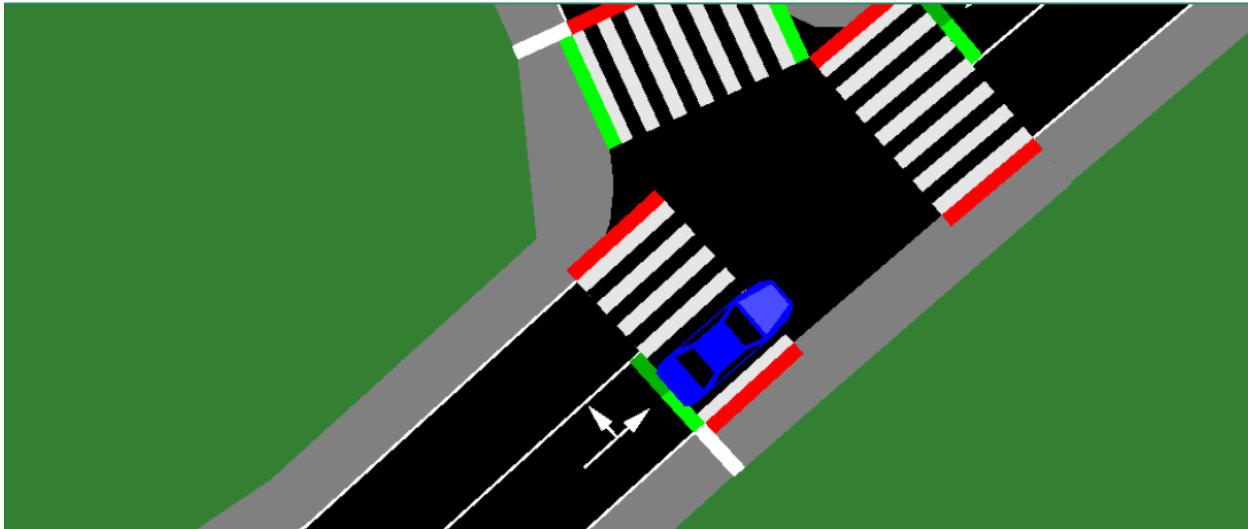
Figure 5.8.1 shows an intersection where an ambulance is approaching and given priority through green signals. Other directions are halted with red signals to clear the path. A regular vehicle is stopped, ensuring safe passage.



**Figure 5.4: Pedestrian Crossing**



Figure 5.8.2 illustrates vehicles waiting in a queue at a red signal near a pedestrian crossing. The stopped cars indicate controlled traffic flow during non-priority phases. The system ensures vehicles halt before the crosswalk for safety. It reflects proper signal compliance and queue formation.



**Figure 5.5: Smooth vehicle flow**

Figure 5.8.3 shows a vehicle moving through the intersection under a green signal. Other directions remain restricted with red signals to avoid conflicts. The intersection is managed to allow smooth and safe vehicle movement. Pedestrian crossings are clearly marked and coordinated with signals.

# CONCLUSION

This project successfully developed and validated a reinforcement learning-based traffic signal control system with several key contributions, including a 684-dimensional directional state representation capturing vehicle queues, pedestrian waits and ambulance proximity separately for each approach across 114 junctions; a mathematically formulated multi-objective reward function balancing vehicle waiting, pedestrian waiting, queue length and emergency priority (+100/-500); an active route clearing mechanism that commands lead vehicles to change lanes and accelerate, physically opening a corridor for ambulances; an emergency override system implementing distance-based pre-emption logic forcing green phases for ambulances within 250 meters; and a scalable Gymnasium-compliant architecture supporting parallelized training across 8 SUMO instances.

The experimental results demonstrate:

- 60.2% reduction in vehicle wait time compared to fixed-time control
- 95.6% reduction in ambulance delay (from 27.3 seconds to 1.2 seconds)
- 58.0% reduction in pedestrian wait time
- 76% of ambulance trips achieved zero stops at intersections

The following limitations are acknowledged:

- Simulation-to-Reality Gap: Real-world deployment would face challenges including sensor noise, communication latency and driver behavior unpredictability not modeled in SUMO.
- Computational Requirements: Training required approximately 4 hours on 8 CPU cores, which is feasible but may require optimization for larger networks.
- Single Emergency Vehicle Assumption: The system prioritizes only one ambulance at a time; simultaneous emergencies could cause conflicts.
- No Weather or Incident Modeling: Adverse weather conditions and traffic incidents are not currently incorporated into the state space.
- Fixed Junction Topology: The system assumes static junction geometry and lane configurations.

## FUTURE SCOPE

- **V2X Integration:** Vehicle-to-Everything (V2X) communication would enable direct data exchange between ambulances and traffic controllers, reducing reliance on simulated radar and providing more accurate position and intent information.
- **Predictive Analytics:** Incorporating historical traffic pattern analysis would allow the system to anticipate congestion before it forms, proactively adjusting signal timing rather than reacting to existing queues.
- **Multi-City Scaling:** Adapting the model architecture and training pipeline to handle diverse city layouts (grid, radial, organic) would increase generalizability. Transfer learning could accelerate deployment to new cities.
- **Real-World Deployment Considerations:** Future work should address hardware requirements for real-time inference (sub-100ms latency), fail-safe mechanisms for communication loss, integration with existing SCADA and traffic management systems and regulatory compliance and safety certification.
- **Multi-Agent Reinforcement Learning (MARL):** Extending to MARL where each junction has its own policy could improve scalability to networks exceeding 500 junctions, though coordination mechanisms would be required.
- **Mixed Emergency Vehicle Types:** Supporting fire trucks (larger turning radius, higher priority) and police vehicles (variable priority based on incident type) would increase real-world utility.

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